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PI Name:	Talley, Douglas G Ph.D.		
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PI Email:	Douglas.talley@edwards.af.mil	**Fax:**	FY 661-275-6245
PI Organization Type:	GOVERNMENT	**Phone:**	661-275-6174
Organization Name:	Air Force Research Laboratory		
PI Address 1:	AFRL/PRSA		
PI Address 2:	10 East Saturn Boulevard		
PI Web Page:			
City:	Edwards AFB	**State:**	CA
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Contact Email:	walter.m.duval@nasa.gov		
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Task Description:	The objective of this research is to develop and experimentally validate a near-critical transcritical and supercritical fluid shear flow model independent of turbulence. We define a supercritical shear flow to be one in which all of the fluid particles remain above their critical temperature and pressure. We define a transcritical shear flow to be one in which at least some of the fluid particles undergo a transition between a subcritical and a supercritical temperature, between a subcritical and a supercritical pressure, or both. The reason it is necessary to validate the fluid model independent of turbulence is that turbulence introduces a large number of additional mechanisms the understanding of which is embryonic at best for near-critical transcritical and supercritical flows. Validating the fluid model without turbulence uncertainties therefore requires laminar flows. However, laminar flows that are not influenced by gravity are difficult to produce in normal gravity due to the large density gradients involved. Therefore, microgravity experiments are necessary. Experiments consisting of visualizing perturbed and un-perturbed laminar jets in microgravity will be conducted in the ground-based facilities at NASA GRC. Direct numerical simulations of the experiments will be performed at the Jet Propulsion Laboratory. Perform high speed visualizations of perturbed and unperturbed laminar, round, cryogenic jets of liquid nitrogen injected into room temperature gaseous nitrogen and gaseous helium in microgravity at various supercritical pressures. Ensure the experiments are performed such that all initial and boundary conditions are known. Validate a fluid shear model by performing direct numerical simulations of the experiments. Injection processes involving transcritical and supercritical shear flows occur in a number of applications where high pressures contribute to performance benefits. Examples include liquid rocket engines, gas turbines, internal combustion engines, and chemical processing applications. Such shear flows are characterized by very steep density gradients approaching gas/liquid systems, but without surface discontinuities, and require sophisticated treatments of transport properties and the equation of state. A scientific understanding of the physics of these flows does not exist. This will be the first time model containing all the necessary physics will have been directly compared to experimental data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	Introduction Presently, the direction of almost all power and thrust producing devices is to increase the operating chamber pressure. The rationale for this increase in chamber pressure is gains in efficiency and thrust output. As chamber pressure is increased, the fluids inside the injector or combustion chamber may no longer remain below the thermodynamic critical point and become supercritical. At pressures much lower than the critical pres-sure, the liquid breaks up into ligaments and drops, leading to spray combustion. The vast majority of what is currently understood about spray combustion applies to these condi-tions, where the gas-to-liquid density ratio (DR) is very small, in the range of 0.002 to 0.03. However, as the combustion chamber pressure increases, the DR becomes larger and can approach unity. A solid fundamental understanding of the processes that occur under supercritical conditions is limited. The objective of this research was to determine the fluid physics governing transport and mixing in non-reacting transcritical and supercritical shear layers. A combined experimental and modeling approach was proposed wherein transport and mixing could be studied. The proposed experimental approach was to produce inertially dominated laminar jets of cryogenic nitrogen injected into room temperature nitrogen in microgravity. The modeling approach considered here was to perform direct numerical simulations (DNS) of the experiments. A drop tower experiment was designed to satisfy the objectives of the program. However, because of the complexities introduced by satisfying safety standards associated with performing drop tower experiments on pressure vessels with windows containing cryogenic fluids at high pressures, it was discovered that the resources required to fabricate the experiment were prohibitive. Consequently, the objectives of the program were modified. On the experimental side, it was decided to expand on supercritical experimental studies already begun in an existing normal gravity facility. Rather than try to produce inertially dominated shear flows in the absence of buoyancy in normal gravity, it was decided instead to study the interaction of subcritical and supercritical shear flows – spe-cifically, shear-coaxial jets – with acoustic waves. This was motivated by a need to un-derstand the physics behind combustion instabilities in liquid rocket engines. Then, since experimental data was not going to be available for comparison with the model, and since adding acoustic capability to the model was beyond the scope of the resources available, the objectives of the modeling effort were modified as well. It was decided to (i) inquire into the numerical issues that occur in DNS of supercritical incompressible flows, such as in the experiment, as compared to equivalent large eddy simulations (LES) in which the small turbulent scales are not computed as in DNS but instead are modeled, (ii) determine the behavior of the small turbulent scales That resolved in DNS but modeled in LES, and (iii) explore what would be the quantity/quantities to measure in an ideal experiment to compare to the numerical solution to ensure that code validation is meaningful. Results from this research program are briefly summarized below. Experimental Summary: The initial task was to design a high pressure experimental chamber to inject a laminar cryogenic jet while the entire test setup is released in the 2.2 sec NASA drop tower. NASA has very stringent safety requirements for chamber operation at pressures of interest to us (2000 psi). To ensure a safe design, an external firm was contracted to perform detailed design of the apparatus. All design calculations were performed by a profes-sional engineer (PE) including a finite element analysis (FEA) of the structure. Details of the FEA analysis are reported in Appendix A. As stated in NASA Technical Memorandum 107090, all components were designed to withstand four times maximum average working pressure (MAWP) – i.e., 8000 psig. Because of the rigid weight constraints, it was necessary to select a material that had a high strength-to-weight ratio to minimize chamber weight and yet not compromise safety. In addition to having a high strength-to-weight ratio, the alloy selected (Ti Grade 23) remains ductile at cryogenic temperatures of interest in this work [1]. Detailed machine shop drawings of the chamber which includes the injection pump for the cryogenic nitrogen have been completed along with a step-by-step details of the design equations and FEA analysis. These are given in Appendix A of the final report. Two companies possessing an ASME Boiler and Pressure Vessel Code were contacted to fabricate the device. The ARB, Inc. Company (Bakersfield, CA) is capable of producing ASME Boiler and Pressure Vessel certified pressure vessels under Section VIII, Division I of the code. However, ARB stated that the apparatus discussed here was beyond the scope of Division I. Another company, Precision Components Corporation (PCC, York, PA), which is certified for Section VIII, Divisions I, II, and III, was able to fabricate an ASME certified vessel under Division II, if the necessary

material for the fabrication was approved by the ASME. However, Ti Grade 23 was not approved by ASME Section VIII, and our calculations showed that no other material could produce a low enough chamber weight to meet the weight limitation and produce meaningful scien-tific data. PCC also stated that it was capable of producing this chamber in an “ASME-like” manner, which would include following the ASME code, except for the material, but including testing of the final machined parts. The term “ASME-like” manner means that the guidelines followed for the manufacturing of the chamber are those recom-mended by the ASME codes and is built by a machine shop possessing ASME section VIII division 1 or 2 code stamp, with the exception of the selected material. Considering that this chamber was intended to withstand four times the MAWP, that it had to be made from the highest strength-to-weight ratio titanium alloy to meet the weight limitation, and had internal dimensions less than the required values by the NASA Glenn Safety Manual, and that it had to be manufactured in an “ASME-like” manner, we consequently wrote a memo to NASA and requested reconsideration of the ASME Code Stamp requirement for this project.

More importantly, it was projected by PCC that the total cost of the “ASME-like” fabrication would be greater than $100,000. This set a budgetary limitation, and was greater than our level of available funds at that time by at least $50,000. NASA was approached for additional funding which proved not to be available. Therefore, the experimental work continued in an existing normal gravity facility, where the emphasis was changed to understanding the interaction between acoustic waves and shear-coaxial jets.

To gain a better understanding of some of the underlying physics associated with the interaction of high-amplitude acoustic waves with a coaxial-jet injector similar to those used in cryogenic liquid rockets, a non-reacting-flow experimental investigation was conducted under sub-, near-, and supercritical chamber pressures, with and without acousti-cal excitation. The flow from the inner jet of this coaxial injector was liquid nitrogen (or liquid-like, if at supercritical pressures) and cold gaseous nitrogen flowed from the outer annular jet, both injected into a chamber pressurized with nitrogen at room temperature. The jet was excited with a high-amplitude acoustic driver, with the jet located at a velocity anti-node (pressure node) of the established acoustic field. Past research on this subject has shown both the relevance and importance of geometrical changes in an injector’s exit-area and its nearby physical and fluid mechanical processes. On this basis, special attention was paid to collecting spatially-resolved mean temperatures and documenting the aforementioned interactions at the exit of this injector. Short-duration and high-speed digital photography, at framing rates up to 18 kHz, provided information on the dynamic behavior of this jet under a variety of conditions. Mean and root mean square (RMS) values of the coaxial-jet dark-core length fluctuations were measured from the acquired images via a computer-automated method. The conclusions of this work are summarized below. Additional details may be found in refs. [2-8].

Reference [2] is reproduced in its entirety in Appendix B of the final report (Ed. note: not available to the public; contact PI).

1. The existence of high-amplitude acoustic waves alters the behavior of the shear-coaxial jet. The resulting structure of the jet exhibits a periodic shape correspond-ing to the transverse-velocity field created by the acoustic waves. The periodic oscillations imposed on the dark core of the jet is predominately in the direction of the transverse acoustic velocity. No helical mode for the jet was observed.

2. The root mean square (RMS) variation of the dark-core length decreases with in-creasing outer-to-inner-jet velocity ratio at a given chamber pressure and asymp-totically approaches a constant level. The root mean square (RMS) of the dark-core length is greatest at subcritical pressures. To the best of the authors’ knowledge, this behavior has not previously been reported in the literature.

3. Previous research showed that an episode of so-called “temperature ramping”, used for rocket combustion stability rating, could lead the engine to an unstable behavior. Also, from other work, it has been shown that coaxial injectors with high outer-to-inner jet velocity ratios (greater than ~10 for LOX/H2 engines) are more stable. In the current work, the observation that the RMS of the dark-core length fluctuations decreased at high velocity ratio, is considered to be a potential explanation for the temperature ramping effect. Reducing the hydrogen tempera-ture increases its density and therefore reduces its velocity. It is possible that the increases in the RMS fluctuations this causes enables a key feedback mechanism for the self-excitation process that drives combustion instability in liquid rocket engines. Ultimately, this hypothesis requires further testing in a multi-element, fired rocket experimental facility, for validation purposes.

4. The quantitative behavior of the dark-core length of the coaxial jet at near- and supercritical pressures follows a similar momentum flux ratio (M) dependency reported for the single-phase shear-coaxial jets (i.e., 12M -0.5). The dark-core length for the subcritical chamber pressures, however, scales with M -0.2.

Task Progress:

5. Within a range of momentum flux ratios in between 1 and 10, the dark-core lengths for the coaxial jet under the subcritical (two-phase) ambient pressure used here appear longer than those measured at the supercritical (single-phase) condi-tion. This range represents relevant values for liquid rocket engines. The subcritical data reported herein was conducted in parameter ranges that have not been re-ported previously.

Numerical Simulation Summary:

The modeling and simulation tasks were as follows. 1. At the very low Mach number of the experiments, it was expected that the acoustic waves have little impact on the flow phenomena of interest, and only flow develop-ments on the scale of the flow velocity need to be resolved. The goal was to obtain an algorithm where the time step is constrained by the flow velocity and not by the acoustic speed. Because we wished to perform computations at Mach number values of 5 10-5, we first conducted an extensive literature search for low-Mach number algorithms, either applicable to or that can be adapted to, the present study. As a result, we implemented a variety of low-Mach number schemes. The low-Mach number schemes that have been considered include computing the pressure implicitly [9, 10], and preconditioning with pseudo-time integration [11]; these schemes involved modifying the numerical scheme while retaining the original governing equations. Our studies determined that neither approach was successful in the supercritical setting where real-gas equations of state must be used, and where we know that there are regions of high-density-gradient magnitude which must be solved in DNS. In contrast, no such requirement is imposed on LES, which use presumed submodels; in LES pre-conditioning techniques are possible. Therefore we explored an alternative approach based on low Mach number asymptotics [12,13]. The goal was to modify the governing equations of the compressible for-mulation to eliminate the acoustic waves while incorporating physically reasonable assumptions to allow the governing equations to be solved with larger physical time steps than those possible for the original governing equations. This is the advantage of the low-Mach-number asymptotics, which we formally derived: the modified gov-erning equations (“low-Mach-number formulation”) can be solved at larger time-steps than required by the original equations. The

scheme derived solves the original governing equations (“compressible formulation”), by integrating different variables, in a numerically stable manner at low Mach numbers. Whereas the assumptions used to derive the low-Mach-number formulation from the compressible formulation were verified to be reasonable based on simulations using the compressible formulation, the low-Mach-number formulation did not yield physically consistent results with the compressible formulation. The lack of agreement was attributed to failure of the low-Mach-number formulation to properly represent the pressure field. Therefore at this juncture, it appears that the low-Mach-number approach is not feasible for this problem, and such calculations should proceed with the compressible formulation. Because for all low-Mach-simulations it was observed that the results at the lowest rea-sonable Mach number for the compressible formulation (0.05) were very similar to those at the target experimental Mach number (5 10-5), except for the pressure field which scales as Mach number squared, the indications are that the Mach=0.05 results will be generally applicable to the lower Mach numbers. The success of these studies was to show that existing numerical methodologies for LES are not applicable to DNS for high-pressure incompressible flows. Since the LES equations are obtained from filtering the DNS equations, thereafter making modeling assumptions neglecting several terms obtained through the filtering of the DNS equations, and then hypothesizing the mathematical form of the small-scale fluxes, it is appropriate to question the validity of the LES approximations made so far in the literature, particularly because the corresponding DNS results for incompressible flows are not yet available for comparison. With the future advent of numerical techniques applicable to high-pressure, incompressible flow DNS, comparisons between experimental results and simulations will be performed, at which point the issue of LES fidelity compared to DNS may be revisited.

2. To understand the characteristics of supercritical flows, with focus on the differences between DNS and LES discussed above, a study was conducted examining the small-scale features of such flows from DNS. Using an existing DNS database of supercritical binary-species temporal mixing layer simulations, explicit mathematical forms (i.e. models) for the small-scale fluxes and simplifying assumptions for the remaining filtered terms were assessed. Coefficients of proportionality calibrated on the DNS database yielded good quantitative agreement with the small-scale fluxes for two of the three types of mathematical forms; however, comparison among the several realizations in the DNS database revealed that, statistically speaking, the calibrated coefficients were not generally valid. The study has been presented in detail in [14]. This shows the complexity of the small turbulent scales behavior that is different under su-percritical conditions from equivalent results obtained under subcritical conditions [15]. Additionally, the implication is that the coefficients cannot be considered con-stant during LES calculations.

3. In the goal of future objective comparisons between ideal experiments and simulations we have focused on the necessary identification of vertical structures as a criterion for comparing experiments and simulations; this is a necessary, but not a sufficient condition for model validation. We have obtained a quantitative criterion to distinguish vortical structures in images, such as would be provided by ideal experiments, and showed that unlike other criteria proposed for incompressible low-pressure flows, the selected criterion can identify the coherent structures expected from high-pressure experiments. A manuscript has been submitted for publication [16].

References [14,16] are reproduced in Appendix C of the final report (Ed. note: not available in public Task Book report; contact PI).

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Articles in Peer-reviewed Journals	Okong'o N, Bellan J. "Vortex identification in high-pressure flows." Physics of Fluids (submitted), August 2005 , Aug-2005
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Papers from Meeting Proceedings	Davis DW, Chehroudi B. "Shear-Coaxial Jets from a Rocket-Like Injector in a Transverse Acoustic Field at High Pressures." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 9 – 12, 2006 AIAA paper AIAA-2006-758, 2006. , Jan-2006
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